Khang Luong

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Research Objective

Computational physicist specializing in quantum scattering theory and ultrafast molecular dynamics, with direct expertise in atomistic simulation of femtosecond electron diffraction—the foundational technique driving AMO research at SLAC MeV-UED and Los Alamos LANSCE. Seeking a research position to extend physics-informed machine learning methods to structure retrieval from time-resolved diffraction data.

Education

University of Nebraska–Lincoln — *Lincoln, NE*

B.S. Physics *Minor: Mathematics & Computer Science* *GPA:* ***3.97 / 4.00*** — *Expected May 2028*

Relevant Coursework: Modern Physics, Quantum Mechanics, Differential Equations, Data Structures & Algorithms, Scientific Computing (Python)

Research Experience

Computational Research Intern — Molecular Excitons & Multiscale Simulation — *Summer 2025*

Shao Lab, Department of Chemistry, Brandeis University — *Waltham, MA*

- Stress-tested and optimized the Shao Lab's Python-based electronic-structure code for modeling molecular excitons and excitonic polaritons—quantum optical states at the light–matter coupling boundary central to next-generation photovoltaic, quantum information, and cavity QED applications.
- Benchmarked the lab's exciton dynamics code for energy-transport simulation across molecular aggregates, systematically probing the accuracy–cost tradeoffs in physics- and ML-based multiscale methods that underpin the group's simulation stack.
- Authored software documentation and reproducible tutorials for both codebases, reducing onboarding friction for incoming researchers and establishing a maintainable standard for simulation workflow documentation across the group.
- Presented research progress in daily meetings with PI Dr. Yihan Shao and postdoc Dr. Zheng Pei, operating at the full accountability cadence of a professional research group throughout the summer.

Undergraduate Research Assistant — *September 2024 – Present*

Department of Physics & Astronomy, University of Nebraska–Lincoln

- Engineered a Python simulation framework implementing ionized atomic form factors within the Independent Atom Model (IAM) for ultrafast electron scattering, reproducing femtosecond-pulse diffraction patterns with residuals quantitatively consistent with experimental measurement uncertainty—directly enabling structural retrieval of transient molecular geometries.
- Derived quantum mechanical elastic scattering amplitudes for multiply ionized molecular species by integrating charge-state-dependent electron density distributions, extending the standard neutral-atom IAM to model nonequilibrium charge states produced in strong-field ionization experiments.
- Constructed a high-fidelity synthetic diffraction data pipeline (NumPy / SciPy) capable of generating theoretical $s\,M(s)$ curves across a configurable momentum-transfer range, providing a reproducible benchmark tool for validating new experimental geometries at the UNL ultrafast electron scattering apparatus.

Research Projects

Computational Modeling of Ionized Molecular Diffraction — *UNL Physics & Astronomy* — *2025 – Present*

- Architected a modular scattering-analysis codebase separating form-factor calculation, molecular geometry specification, and diffraction pattern generation, reducing experiment-to-simulation turnaround from hours to under five minutes for new molecular targets.
- Implemented vectorized numerical integration of atomic scattering factors over a discrete momentum-transfer grid using SciPy `quad` and NumPy broadcasting, achieving a **40**$\times$ speedup over the original loop-based prototype while maintaining sub-0.1% numerical precision.
- Produced publication-quality $s\,M(s)$ and pair distribution function (PDF) plots in Matplotlib used directly in group presentations, providing quantitative comparison between theory and femtosecond electron pulse experimental results.

Physics-Informed CNN for Scientific Image Classification — *Hackathon Project* *Fall 2025*

- Designed and trained a convolutional neural network (TensorFlow / Keras) on a **13,000-image** dataset, achieving robust multi-class classification with uncertainty estimation via Monte Carlo dropout— demonstrating direct applicability to diffraction pattern recognition and anomaly detection in high-throughput AMO experiments.
- Deployed a full-stack inference pipeline (React front end, Python REST back end) within a **24-hour** constraint, validating the model's readiness for real-time data acquisition integration of the kind required at LCLS and LANSCE beam-time allocations.

Honors & Awards

- **UCARE Research Fellowship**, University of Nebraska–Lincoln 2025 – Present
  Competitive undergraduate research grant; awarded to <10% of applicants university-wide.
- **College of Arts & Sciences Dean's List** (GPA $\geq$ 3.75) 2024 – 2025

Technical Skills

Scientific Computing	Python (NumPy, SciPy, Matplotlib), vectorized numerical integration, ODE/PDE solvers, FFT analysis
Machine Learning	TensorFlow / Keras (CNN, MC dropout), scikit-learn, physics-informed ML, multiscale ML methods
AMO / Quantum Sim.	Quantum scattering theory, IAM & ionized form factors, ultrafast electron diffraction (GUED/UED), pair distribution functions, molecular exciton & excitonic polariton modeling, exciton transport, excited-state dynamics, cavity QED concepts
Software Engineering	Git, modular Python package design, scientific software testing & documentation, REST API development, React / Vite, LaTeX
Languages	English (fluent), Vietnamese (native)

Physics & Mathematics Foundation

Modern Physics (Special Relativity, QM intro)	Differential Equations (analytical & numerical)	Data Structures & Algorithms
Scientific Computing with Python	Calculus I–III	Linear Algebra

Currently enrolled: Introduction to Electrodynamics, Electrical and Electronic Circuit